

Project Initialization and Planning Phase

Date	30 September 2026
Team ID	SWTID-2026-1758
Project Title	EduGenie – AI Powered Learning Assistant
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to enhance education through Artificial Intelligence by providing personalized learning support, instant doubt resolution, concept explanations, quiz generation, and content summarization. The solution improves learning efficiency, engagement, and academic performance using Google Gemini AI.

Project Overview	
Objective	The primary objective of EduGenie is to provide an AI-powered learning assistant that helps students understand concepts, generate quizzes, summarize content, and receive personalized learning support.
Scope	The project focuses on enhancing the learning experience through artificial intelligence by providing instant doubt resolution, concept explanations, quiz generation, and personalized study assistance.
Problem Statement	
Description	Students often face difficulties in understanding complex concepts, finding relevant learning resources, and receiving personalized guidance. Traditional learning methods are time-consuming and may not meet individual learning needs.
Impact	These challenges can reduce learning efficiency, increase frustration, and negatively affect academic performance. Addressing these issues can improve learning outcomes and student engagement.
Proposed Solution	
Approach	The solution uses Google Gemini AI and FastAPI to deliver intelligent educational assistance through concept explanations, quiz generation, text summarization, and personalized learning recommendations.
Key Features	AI-powered Question Answering Concept Explanation, Quiz Generation, Text Summarization, Personalized Learning Path

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Intel/AMD Processor, Multi-Core CPU
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	1 TB SSD
Software		
Frameworks	Python frameworks	FastAPI
Libraries	Additional libraries	google-generativeai, requests, python-dotenv
Development Environment	IDE	HTML, CSS, JavaScript
Data		
Data	Source, size, format	Google Gemini API